Ш of $y^2 = x^3 + px$ at the Fermat primes

Why the shape $(\mathbb{Z}/2^{k-1})^2$ is forced, and what the pattern really asks

on a question of Rene Pannekoek

1. The observation

For $E_d : y^2 = x^3 + dx$ and $p = 2^{2^k} + 1$ a Fermat prime, computation suggests

$$Ш(E_p) \cong (\mathbb{Z}/2^{k-1}\mathbb{Z})^2,$$

that is $Ш = 0, (\mathbb{Z}/2)^2, (\mathbb{Z}/4)^2, (\mathbb{Z}/8)^2$ for $p = 5, 17, 257, 65537$.

The pattern splits cleanly into two questions of very different character, and this is the main point of what follows:

(a) Why is Ш of the form $(\mathbb{Z}/2^e)^2$? — because it cannot be anything else. This is forced, needs no BSD, and is proved in Section 2.

(b) Why is $e = k - 1$? — this is the whole content, and it is equivalent to a single statement about an L-value, Section 3. We do not explain it.

2. The shape is forced

Theorem 1. Let $p \equiv 1 \pmod 8$ be a prime with $L(E_p, 1) \neq 0$. Then

$$Ш(E_p)[2^\infty] \cong (\mathbb{Z}/2^e\mathbb{Z})^2 \quad \text{for some } e \geq 1.$$

No conjecture is used.

Proof. Four inputs.

(i) E_p has complex multiplication by $\mathbb{Z}[i]$, so by Coates–Wiles $L(E_p, 1) \neq 0$ forces rank $E_p(\mathbb{Q}) = 0$, and by Rubin $Ш(E_p)$ is finite.

(ii) $E_p(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/2$, generated by $(0, 0)$: the curve $y^2 = x^3 + dx$ has full 2-torsion only when $-d$ is a square, and $\mathbb{Z}/4$ only for $d = 4$.

(iii) $\mathrm{Sel}^{(2)}(E_p) \cong (\mathbb{Z}/2)^3$ for $p \equiv 1 \pmod 8$ — this is the computation in Silverman X.6.

(iv) The descent sequence

$$0 \to E_p(\mathbb{Q})/2E_p(\mathbb{Q}) \to \mathrm{Sel}^{(2)}(E_p) \to Ш(E_p)[2] \to 0$$

now reads $0 \to \mathbb{Z}/2 \to (\mathbb{Z}/2)^3 \to Ш[2] \to 0$, since rank 0 and torsion $\mathbb{Z}/2$ give $\dim E_p(\mathbb{Q})/2 = 1$. Hence

$$\dim_{\mathbb{F}_2} Ш(E_p)[2] = 2.$$

A finite abelian 2-group with $\dim A[2] = 2$ is a product of **exactly two** cyclic groups, $\mathbb{Z}/2^{e_1} \times \mathbb{Z}/2^{e_2}$; and the Cassels–Tate pairing on Ш is alternating and non-degenerate, which forces the 2-primary part to be of the form $A \times A$, so $e_1 = e_2$. ∎

So **nothing** about the Fermat primes is needed for the shape: every $p \equiv 1 \pmod 8$ with $L(E_p, 1) \neq 0$ has $Ш[2^\infty] = (\mathbb{Z}/2^e)^2$. Two independent confirmations from the computation:

- for $p = 17, 257, 65537$, `ellrank` returns the interval $0..2$ rather than a rank — and PARI's documentation says the 2-descent "never" determines the rank when Ш has 4-torsion. It is telling us, without being asked, that Ш has 4-torsion for $p = 257$ and 65537 and not for $p = 17$;
- $\dim \mathrm{Sel}^{(2)}$ comes out 3 for $p = 17, 257, 65537$ and 2 for $p = 5$, matching (iii).

3. The size is one L-value

Now bring in BSD. For every one of the 64 primes $p \equiv 1 \pmod 8$ below 3000 with $L(E_p, 1) \neq 0$ we computed

$$\text{the Tamagawa product } c = 2, \quad \text{the torsion } |T| = 2,$$

so the BSD formula collapses to

$$\#Ш(E_p) = \frac{L(E_p, 1)}{\Omega} \cdot \frac{|T|^2}{c} = 2 \cdot \frac{L(E_p, 1)}{\Omega}.$$

And in all 64 cases the L-value has the **Waldspurger square shape**

$$\frac{L(E_p, 1)}{\Omega} = 2m^2, \quad m \in \mathbb{Z}_{>0},$$

so that $\#Ш = (2m)^2$ — consistent with Theorem 1, and with the odd part of Ш being $(\mathbb{Z}/m_{\text{odd}})^2$. Writing $e = 1 + v_2(m)$ we get $Ш[2^\infty] = (\mathbb{Z}/2^e)^2$.

So the question is exactly: why is $m(p) = 2^{k-2}$, equivalently

$$\frac{L(E_p, 1)}{\Omega} = 2^{2k-3},$$

for $p = 2^{2^k} + 1$?

The three computable cases, the last of which we computed here rather than taking on trust:

k	p	$L(1)/\Omega$	m	$\#Ш$	Ш
1	5	— (rank 1)	—	1	0
2	17	2	1	4	$(\mathbb{Z}/2)^2$
3	257	8	2	16	$(\mathbb{Z}/4)^2$
4	65537	**32**	4	**64**	$(\mathbb{Z}/8)^2$

4. What is special about a Fermat prime here

Two things, and it is not clear which is doing the work.

The Gaussian factorisation is as degenerate as possible. Writing $p = a^2 + b^2$ with a odd,

$$p = 2^{2^k} + 1 = 1^2 + \left(2^{2^{k-1}}\right)^2 \quad \Longrightarrow \quad a = 1, \quad b = 2^{2^{k-1}}.$$

So a is as small as it can be and b is a pure power of 2 — the splitting $p = \pi\overline{\pi}$ in $\mathbb{Z}[i]$ has $\pi = 1 + 2^{2^{k-1}} i$.

The unit group is a 2-group. $\#\mathbb{F}_p^\times = p - 1 = 2^{2^k}$, so every quartic, octic and higher residue symbol modulo p takes values in 2-power roots of unity, and 2 has order exactly 2^{k+1} modulo p, since $2^{2^k} \equiv -1$.

Either could plausibly force the 2-adic valuation of the L-value. But the data warn against the easy guess that $v_2(b)$ alone decides m:

p	a	b	m
4177	9	64	2
4217	11	64	2

p	a	b	m
4721	25	64	2
4937	29	64	**4**

Same b, same $v_2(b) = 6$, and m jumps. Whatever governs m sees more than the 2-adic size of b. (Nor is m always a power of 2: $p = 857$ gives $m = 3$ and $\#Ш = 36$, and $p = 2017, 2617$ also give $m = 3$.)

5. Two remarks on the question as posed

The weaker question is not weaker. Since Theorem 1 gives $Ш[2^\infty] = (\mathbb{Z}/2^e)^2$, the **exponent** of $Ш[2^\infty]$ is exactly 2^e. So "is the exponent divisible by 2^{k-1}" and "is $Ш \cong \left(\mathbb{Z}/2^{k-1}\right)^2$" are the same statement, once the shape is known. There is no intermediate target to aim at.

Why $d = 2^{32} + 1$ hangs. That d is composite, so Silverman X.6 does not apply and even the Selmer rank is not given by the same recipe. Worse, the conductor is $2^6 d^2 \approx 1.2 \cdot 10^{21}$, so an L-value computation needs on the order of $\sqrt{N} \approx 3.4 \cdot 10^{10}$ terms. That is why Sage and Magma do not return: it is not a subtlety, just the size.

6. Status

Proved, unconditionally: Theorem 1. For every prime $p \equiv 1 \pmod 8$ with $L\big(E_p, 1\big) \neq 0$, $Ш\big(E_p\big)[2^\infty] \cong (\mathbb{Z}/2^e)^2$. This answers the shape half of the question and removes the mystery from it — the pattern's most striking feature, that Ш is a **square** of a cyclic group, is a theorem about any such p, not a property of Fermat primes.

Computed: $c = |T| = 2$ and $L(1)/\Omega = 2m^2$ for all 64 rank-0 primes $p \equiv 1 \pmod 8$ below 3000; and $L(E_{65537}, 1)/\Omega = 32$, giving $\#Ш = 64$ — the $k = 4$ case, verified here independently.

Not explained: why $m = 2^{k-2}$ at the Fermat primes. Everything reduces to that one statement, and it is a statement about the 2-adic valuation of a Waldspurger coefficient in the quadratic twist family $y^2 = x^3 + dx$. The natural place to look is a Tunnell-style formula for that family, with the Fermat condition entering through $a = 1$, $b = 2^{2^{k-1}}$.